

3. Research Contribution

What Was Explored

The project explored practical deepfake detection using convolutional neural networks and explainable AI techniques. Particular attention was given to image classification, frame-level video analysis, model explainability, and deployment considerations.

The work also investigated how synthetic media intersects with cybersecurity threats such as identity impersonation, misinformation campaigns, social engineering, and fraud.

What Was Learned

Key learnings include:

- Deepfake detection is highly dependent on dataset quality.
- High validation accuracy does not guarantee real-world robustness.
- Explainability techniques improve transparency and user trust.
- Video analysis requires temporal consistency and aggregation methods.
- Mobile deployment introduces engineering challenges beyond model training.

Cybersecurity Relevance

Deepfakes represent an emerging cybersecurity threat because they can be weaponized to manipulate public opinion, impersonate individuals, bypass trust mechanisms, and facilitate fraud.

The project connects machine learning research with cybersecurity applications by focusing on media authenticity and digital trust.

Future Research Opportunities

Potential future directions include:

- Cross-dataset generalization studies
- Multimodal misinformation detection
- Transformer-based architectures
- Adversarial robustness

- Media provenance verification
- Explainable AI evaluation
- Deepfake detection for Indian languages and demographics

The project serves as a foundation for further exploration of AI-driven cybersecurity solutions.